

APM1110: Probability and Probability Distributions

Formative Assessment 5 — Solutions (Total: 30 Points)

Problem 1: Discrete Random Variables [7 Points]

A discrete random variable X has PMF:

$$P(X = x) = k(2x + 1), \quad x = 1, 2, 3, 4$$

(a) Find the value of the constant k [2 pts]

Since all probabilities must sum to 1:

$$k[(2(1) + 1) + (2(2) + 1) + (2(3) + 1) + (2(4) + 1)] = 1$$

$$k(3 + 5 + 7 + 9) = 1 \implies k(24) = 1 \implies \boxed{k = \frac{1}{24}}$$

The resulting probability distribution:

x	1	2	3	4
$P(X = x)$	$3/24$	$5/24$	$7/24$	$9/24$

(b) Compute $E[X]$ and $Var(X)$ [3 pts]

$$E[X] = \sum x P(x) = 1 \left(\frac{3}{24}\right) + 2 \left(\frac{5}{24}\right) + 3 \left(\frac{7}{24}\right) + 4 \left(\frac{9}{24}\right) = \frac{3 + 10 + 21 + 36}{24} = \frac{70}{24} = \frac{35}{12} \approx \boxed{2.9167}$$

$$E[X^2] = 1 \left(\frac{3}{24}\right) + 4 \left(\frac{5}{24}\right) + 9 \left(\frac{7}{24}\right) + 16 \left(\frac{9}{24}\right) = \frac{3 + 20 + 63 + 144}{24} = \frac{230}{24} = \frac{115}{12} \approx 9.5833$$

$$Var(X) = E[X^2] - (E[X])^2 = \frac{115}{12} - \left(\frac{35}{12}\right)^2 = \frac{1380 - 1225}{144} = \frac{155}{144} \approx \boxed{1.0764}$$

(c) Calculate $P(X \geq 3)$ [2 pts]

$$P(X \geq 3) = P(3) + P(4) = \frac{7}{24} + \frac{9}{24} = \frac{16}{24} = \frac{2}{3} \approx \boxed{0.6667}$$

Problem 2: Binomial Distribution [7 Points] $X \sim \text{Binomial}(n = 20, p = 0.08)$ **(a) $P(X = 2)$ [2 pts]**

$$P(X = 2) = \binom{20}{2} (0.08)^2 (0.92)^{18} = 190 \times 0.0064 \times 0.222936 \approx \boxed{0.2711}$$

(b) $P(X \geq 3)$ [3 pts]

$$P(X = 0) = (0.92)^{20} \approx 0.1887$$

$$P(X = 1) = \binom{20}{1} (0.08)^1 (0.92)^{19} \approx 0.3282$$

$$P(X = 2) \approx 0.2711$$

$$P(X \geq 3) = 1 - [P(0) + P(1) + P(2)] = 1 - 0.7879 \approx \boxed{0.2121}$$

(c) Mean and standard deviation [2 pts]

$$\mu = np = 20(0.08) = \boxed{1.6000}$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{20(0.08)(0.92)} = \sqrt{1.472} \approx \boxed{1.2133}$$

Problem 3: Poisson Distribution [8 Points] $\lambda = 4.5$ hits per minute**(a) $P(X = 6)$ [2 pts]**

$$P(X = 6) = \frac{e^{-4.5}(4.5)^6}{6!} = \frac{(0.0111090)(8303.7656)}{720} \approx \boxed{0.1281}$$

(b) $P(X \leq 2)$ [3 pts]

$$P(0) = e^{-4.5} \approx 0.011109$$

$$P(1) = e^{-4.5}(4.5) \approx 0.049990$$

$$P(2) = \frac{e^{-4.5}(4.5)^2}{2!} \approx 0.112479$$

$$P(X \leq 2) = 0.011109 + 0.049990 + 0.112479 \approx \boxed{0.1736}$$

(c) $P(X > 10)$ over a 2-minute interval [3 pts]

For a 2-minute interval: $\lambda' = 4.5 \times 2 = 9$, so $X \sim \text{Poisson}(9)$.

k	$P(X = k)$
0	0.000123
1	0.001111
2	0.004998
3	0.014994
4	0.033737
5	0.060727
6	0.091090
7	0.117116
8	0.131756
9	0.131756
10	0.118580

$$P(X \leq 10) = \sum_{k=0}^{10} P(k) \approx 0.7060$$

$$P(X > 10) = 1 - P(X \leq 10) = 1 - 0.7060 \approx \boxed{0.2940}$$

Problem 4: Continuous Probability & Normal Distribution [8 Points]

$X \sim N(\mu = 120, \sigma = 15)$ milliseconds

(a) $P(X > 135)$ [2 pts]

$$Z = \frac{135 - 120}{15} = 1.00$$

$$P(X > 135) = 1 - \Phi(1.00) = 1 - 0.8413 = \boxed{0.1587}$$

(b) $P(100 < X < 130)$ [3 pts]

$$Z_1 = \frac{100 - 120}{15} = -1.3333, \quad Z_2 = \frac{130 - 120}{15} = 0.6667$$

$$\Phi(0.6667) \approx 0.7475, \quad \Phi(-1.3333) \approx 0.0912$$

$$P(100 < X < 130) = 0.7475 - 0.0912 \approx \boxed{0.6563}$$

(c) 95th percentile [3 pts]

The z -score corresponding to the 95th percentile is $z_{0.95} = 1.6449$.

$$X_{0.95} = \mu + z\sigma = 120 + 1.6449(15) = 120 + 24.6735 \approx \boxed{144.6735 \text{ ms}}$$

Prepared for APM1110: Probability and Probability Distributions — Formative Assessment 5